

MECHSOURCE

DOCUMENT 02

RESEARCH, DATA AND BUSINESS REQUIREMENTS

Product MECHSOURCE

Document 02 Research, Data and Business Requirements

Purpose Establish the real world information, datasets, data sources, business processes, verification systems, market conditions and external dependencies required to make MECHSOURCE reliable, useful and scalable.

1. PURPOSE OF THIS DOCUMENT

MECHSOURCE depends on more than software.

The platform will only be useful if it has reliable information about vehicles, equipment, parts, compatibility, suppliers, technicians, workshops, prices, availability and locations.

This document defines what must be researched and established before the final UI and technical architecture are approved.

The objective is to answer

- What information does MECHSOURCE need?
- Where will the information come from?
- Who owns the information?
- Can MECHSOURCE legally use it?
- How will it be verified?
- How frequently will it change?
- Who will maintain it?
- How will it be presented to users?

2. INITIAL MARKET SCOPE

The initial geographical focus is

- Lagos
- Ogun
- Abuja

The platform should eventually expand throughout Nigeria, then West Africa, Africa and international markets.

Research should therefore distinguish between

Initial Nigerian requirements
Future African requirements
International requirements

The first research phase should concentrate on Nigeria without designing the underlying data model in a way that prevents international expansion.

3. MARKET RESEARCH

Research must establish the actual size and behaviour of the target market.

We need to investigate

1. Number and types of vehicles in the target market
2. Common vehicle manufacturers
3. Common vehicle models
4. Common engine types
5. Common commercial vehicles
6. Common truck brands
7. Common diesel engines
8. Common heavy equipment
9. Common construction equipment
10. Common mining equipment
11. Common agricultural equipment
12. Major industrial equipment categories
13. Common spare parts
14. Frequently replaced parts
15. High demand parts

16. High value parts
17. Difficult to source parts
18. Common counterfeit risks
19. Typical customer purchasing behaviour
20. Current methods of finding parts
21. Typical delivery expectations
22. Typical payment preferences
23. Customer trust concerns
24. Supplier concerns
25. Technician concerns
26. Workshop concerns
27. Fleet operator requirements

This research will influence the eventual product priorities and data collection strategy.

4. COMPETITOR RESEARCH

We need to identify existing platforms operating in

- Nigeria
- Africa
- Europe
- North America
- Asia

Research categories should include

1. Automotive parts marketplaces
2. Vehicle marketplaces
3. Heavy equipment marketplaces
4. Industrial marketplaces
5. Equipment rental platforms
6. Mechanic platforms
7. Workshop directories
8. Fleet management platforms
9. Maintenance management platforms
10. B2B procurement platforms
11. AI based parts identification platforms

For each relevant competitor, research should document

Business model
 Target users
 Product categories
 Search experience
 Parts identification process
 Supplier model
 Pricing model
 Delivery model
 Verification model
 Technology approach where publicly available
 Strengths
 Weaknesses
 Customer complaints
 Market positioning
 Potential lessons for MECHSOURCE
 MECHSOURCE should not simply copy existing marketplaces.
 The research should identify gaps that MECHSOURCE can solve better.

5. VEHICLE DATA REQUIREMENTS

The platform will require structured vehicle information.

Potential fields include

- Manufacturer
- Brand
- Model
- Model generation
- Production year
- Production range
- Body type
- Vehicle category
- Fuel type
- Engine capacity
- Engine type
- Engine code
- Transmission
- Drive type
- Market or region
- Vehicle identification information where appropriate

The data structure must allow different vehicles with similar names to remain distinguishable.

6. EQUIPMENT DATA REQUIREMENTS

The same principle must apply to machinery.

Potential equipment data includes

- Manufacturer
- Equipment category
- Model
- Model year
- Generation
- Engine
- Engine code
- Serial information where appropriate
- Operating specifications
- Equipment type
- Industry
- Compatible components
- Parts
- Filters
- Consumables
- Maintenance requirements

The platform should be capable of representing construction, mining, agricultural and industrial equipment.

7. PART DATA REQUIREMENTS

This is one of the most important datasets in the entire project.

MECHSOURCE will require structured part information.

Potential fields include

- Part ID
- Part name
- Part category
- Subcategory
- Manufacturer
- Brand
- OEM number

Manufacturer part number
Alternative part numbers
Cross reference numbers
Description
Specifications
Dimensions where relevant
Material where relevant
Vehicle compatibility
Equipment compatibility
Engine compatibility
Images
Technical documents where legally available
Warranty information
Certification information where relevant
Installation information where legally available
Supplier information
Stock information
Price
Location
Availability

8. OEM AND PART NUMBER DATA

The platform should support multiple identification numbers for the same part.

For example

OEM number
Manufacturer number
Supplier number
Alternative number
Cross reference number

The system should distinguish between these identifiers instead of treating every number as the same type.

This is essential for the Find My Part system.

9. COMPATIBILITY DATA

Compatibility will be a central MECHSOURCE dataset.

The system needs relationships such as

Vehicle → Engine → Part

Vehicle → Part

Equipment → Engine → Part

Equipment → Part

OEM Part → Alternative Part

Manufacturer Part → Cross Reference

The platform should never assume compatibility merely because two product names appear similar.

Compatibility information must have a source and verification status.

Possible statuses include

- Verified
- Supplier supplied
- Technician verified
- Manufacturer sourced
- AI suggested
- Unverified

This distinction should eventually appear in the user experience where necessary.

10. PART DATA SOURCES

Potential sources need to be researched individually.

Possible sources include

- Vehicle manufacturers
- Equipment manufacturers
- Authorised distributors
- Parts manufacturers
- Authorised dealers
- Licensed automotive databases
- Licensed aftermarket catalogues
- Parts distributors
- Verified suppliers
- Workshops
- Technicians
- Publicly available datasets
- Government information
- Industry associations
- Commercial data providers
- AI assisted classification

However, every source must be assessed for

Accuracy

Coverage

Update frequency

Licensing

Commercial usage rights

Cost

API availability

Data format

Geographical coverage

The platform must not assume that information found online is automatically available for commercial reuse.

11. NIGERIAN VEHICLE DATA

Official Nigerian vehicle registration information demonstrates the type of vehicle attributes that can exist in a structured vehicle record.

The Federal Road Safety Corps vehicle registration system includes fields such as vehicle category, make, model, year of manufacture, fuel type, engine number, chassis number and engine capacity.

This does not mean MECHSOURCE should simply copy government registration data.

Instead, it establishes useful categories that should be considered when designing the MECHSOURCE vehicle data model.

12. STANDARDS AND PRODUCT QUALITY

MECHSOURCE will need to understand Nigerian standards affecting products sold through the platform.

The Standards Organisation of Nigeria is the national standards body and its responsibilities include standards relating to products, materials, processes and services. Its mechanical standards activities specifically include motor vehicle parts and other mechanical areas.

SON also maintains standards relating to areas such as road vehicles, tyres, agricultural machinery, small tools and cranes and industrial trucks.

Research must therefore establish which product categories on MECHSOURCE may require particular standards, conformity information, certification or other compliance checks.

13. PRODUCT VERIFICATION

The platform should eventually distinguish between

Verified product

Supplier submitted product

Manufacturer supplied product

Technician verified product

AI identified product

Unverified product

Verification requirements should vary according to product category.

For example, a simple accessory may require less verification than a critical mechanical component.

The final verification framework must be established through business, legal and technical research.

14. COUNTERFEIT AND SUBSTANDARD PRODUCT RISK

This is a major business consideration.

Research should investigate

Common counterfeit automotive products

Counterfeit OEM branding

Substandard parts

Fake certifications

Mislabelled products

Used parts sold as new

Incorrect compatibility claims

Supplier fraud

Product substitution

Price manipulation

MECHSOURCE should develop mechanisms to reduce these risks.

SON's MANCAP programme is designed to ensure locally manufactured products conform to relevant Nigerian Industrial Standards before being placed on the market, illustrating why product quality and conformity information may matter to MECHSOURCE's marketplace model.

15. SUPPLIER RESEARCH

We need to understand the supplier ecosystem in the initial target locations.

Research should identify

- Parts dealers
- Distributors
- Importers
- Manufacturers
- Authorised dealers
- Independent dealers
- Industrial suppliers
- Heavy equipment dealers
- Tool suppliers
- PPE suppliers
- Battery suppliers
- Lubricant suppliers
- Filter suppliers
- Equipment suppliers
- Potential logistics partners

The objective is not merely to create a directory.

We need to understand how these businesses currently

- Receive stock
- Track inventory
- Set prices
- Receive customer requests
- Communicate with customers
- Process orders
- Handle payments
- Arrange delivery
- Handle returns
- Handle warranties

16. SUPPLIER ONBOARDING

Research should establish what information a supplier needs to provide.

Potential information includes

- Business name
- Business registration information
- Business address
- Contact information
- Business category
- Product categories
- Operating locations
- Bank or settlement information
- Verification documents
- Business photographs
- Product information
- Inventory
- Pricing
- Delivery options
- Return policy
- Warranty information

The final requirements should be legally validated before implementation.

17. INVENTORY DATA

Supplier inventory is likely to change frequently.

MECHSOURCE needs to determine how inventory will be maintained.

Possible approaches include

- Manual inventory updates
- Spreadsheet upload
- CSV upload
- Supplier dashboard
- API integration
- Point of sale integration
- Inventory management integration
- Automated synchronisation

The platform should eventually support more than one method.

18. PRICE DATA

Price information should include

- Selling price
- Currency

- Unit
- Discount where applicable
- Bulk price
- Business price
- Minimum order quantity
- Availability
- Price update date
- Supplier
- Location

Prices may differ between suppliers, locations and customer types.

The system should therefore avoid treating price as a permanent property of a product.

19. LOCATION DATA

Location information will support

- Supplier discovery
- Workshop discovery
- Technician discovery
- Emergency services
- Delivery
- Equipment rental
- Equipment marketplace

Search, The platform should support

- Country
- State
- City
- Local government area
- Neighbourhood
- Business address
- Coordinates where appropriate
- Service radius
- Delivery zones

The system should not require customers to expose more location information than necessary.

20. TECHNICIAN DATA

Technician information may include

- Name
- Specialisation
- Experience
- Services
- Location
- Service area
- Availability
- Verification status
- Professional credentials where applicable
- Customer rating
- Reviews
- Completed jobs
- Pricing
- Emergency availability

The final verification requirements must be established through research.

21. WORKSHOP DATA

Workshop information may include

- Workshop name
- Business owner
- Location
- Services
- Vehicle categories
- Equipment categories
- Technician specialisations
- Operating hours
- Availability
- Verification
- Ratings
- Reviews
- Contact information
- Service area

The platform should eventually distinguish between verified and unverified workshops.

22. EQUIPMENT DATA

Research must identify the major equipment categories relevant to MECHSOURCE.

Potential categories include

- Construction machinery
- Mining machinery
- Agricultural machinery
- Industrial machinery
- Generators
- Compressors
- Excavators
- Loaders
- Cranes
- Bulldozers
- Heavy trucks
- Specialised machinery

For each category, we need to determine

- Major manufacturers
- Major models
- Common parts
- Common maintenance requirements
- Common suppliers
- Typical prices
- Rental demand
- Purchase demand
- Service requirements

23. LOGISTICS RESEARCH

The platform needs a realistic delivery model.

Research should determine

Available logistics providers

Delivery coverage

Same day delivery

Next day delivery
Interstate delivery
Heavy item transportation
Fragile item transportation
Tracking
Proof of delivery
Delivery pricing
Supplier pickup
Customer pickup

Cash on delivery where legally and commercially appropriate

Returns

Failed deliveries

The final logistics model should not be designed until these realities are understood.

24. PAYMENT RESEARCH

Payment research should establish

Available payment providers

Card payments

Bank transfer

Other appropriate payment methods

Business payments

Marketplace settlement

Refunds

Chargebacks

Supplier payouts

Commission handling

International payment requirements

The final provider selection belongs in the technical architecture stage after business requirements are established.

25. CONSUMER PROTECTION

MECHSOURCE will operate as a marketplace involving consumers and businesses.

Nigeria's Federal Competition and Consumer Protection Commission identifies the Federal Competition and Consumer Protection Act 2018 as the main statute governing consumer protection and competition regulation in Nigeria.

Research must therefore establish MECHSOURCE's responsibilities concerning

Product information

Pricing

Returns

Refunds

Warranties

Misrepresentation

Supplier conduct

Complaints

Dispute resolution

Marketplace responsibilities

Consumer communications

The FCCPC also provides a complaint handling framework, making dispute and complaint management an important consideration for the platform.

26. DATA PROTECTION

Because MECHSOURCE will process customer, supplier, technician and business information, data protection must be investigated before implementation.

Research should establish requirements relating to

Personal information

Location information

Account information

Business information

Payment related information

Images uploaded by users

Vehicle information

Communication records

Analytics

AI processing

Data retention

Data deletion

User consent

Privacy notices

Third party processors

The specific Nigerian data protection requirements must be reviewed against the current applicable law and regulatory guidance before implementation.

27. AI DATA REQUIREMENTS

AI parts identification requires appropriate training or reference data.

Potential data may include

Part photographs

Part categories

Part names

OEM numbers

Manufacturer information

Vehicle compatibility

Equipment compatibility

Technical characteristics

Images from different angles

Images under different lighting conditions

Damaged part examples

Packaging examples

Labels

Serial numbers where appropriate

AI should not be trained or operated on datasets whose commercial usage rights are unclear.

28. AI IDENTIFICATION DATASET

A dedicated MECHSOURCE image dataset may eventually be required.

A dataset could contain

Image

Part category

Part name

Manufacturer

OEM number

Alternative numbers

Vehicle compatibility

Equipment compatibility

Image quality

Image angle

Condition

Verification status

Source

This dataset could eventually improve MECHSOURCE's image identification capability.

29. HUMAN VERIFICATION DATA

AI alone may not be sufficient for difficult parts.

MECHSOURCE may require a human verification network consisting of

Verified mechanics

Parts specialists

Technicians

Supplier experts

Equipment specialists

The platform could use these experts to review uncertain identification requests.

This creates an important combination

AI identification + human expertise + structured parts data.

30. SEARCH DATA

Search must be capable of handling different ways users describe the same product.

For example

Fuel filter

Diesel fuel filter

Hilux fuel filter

Hilux diesel filter

2.8 Hilux fuel filter

OEM number

The underlying search system therefore needs

Synonyms

Alternative names

OEM numbers

Manufacturer numbers

Vehicle relationships

Equipment relationships

Categories

Supplier information

Location information

Natural language interpretation

31. DATA STANDARDISATION

Supplier information will probably arrive in inconsistent formats.

MECHSOURCE must standardise

Product names

Categories

Brands

OEM numbers

Vehicle names

Model names

Engine names

Units

Prices

Locations

Compatibility information

Inventory status

The system should maintain a standard internal structure while allowing suppliers to use their own terminology during onboarding.

32. DATA QUALITY MANAGEMENT

Every important dataset should have

Source

Date collected

Date updated

Verification status

Responsible party

Confidence level where applicable

Version or revision information where necessary

This allows MECHSOURCE to determine whether information is trustworthy and current.

33. DATA UPDATE STRATEGY

Research must determine how often each type of data changes.

For example

Vehicle specifications may change slowly.

Supplier inventory may change rapidly.

Prices may change frequently.

Technician availability may change daily.

Delivery estimates may change continuously.

AI models may change periodically.

The system should therefore not treat all data as equally static.

34. DATA LICENSING

Before using external datasets, MECHSOURCE must establish

Who owns the data

Whether commercial use is allowed

Whether modification is allowed

Whether redistribution is allowed

Whether API access is available

Whether attribution is required

Whether derived datasets can be created

Whether the licence has geographical restrictions

Whether the licence has user limits

This is essential.

A technically excellent dataset cannot safely become a commercial MECHSOURCE dataset if its licence does not permit the intended use.

35. GOVERNMENT AND INDUSTRY SOURCES

Official sources should be prioritised when researching regulatory and standards information.

For example, SON provides access to Nigerian Industrial Standards and maintains information across mechanical, electrical, civil and other technical categories.

Government and industry sources should be evaluated individually for

Authority

Accuracy

Update frequency

Access method

Licence

Commercial usage

API availability

36. BUSINESS MODEL RESEARCH

Before choosing the final revenue structure, research should determine what customers and suppliers are willing to pay for.

Potential revenue models include

Marketplace commission

Supplier subscription

Premium supplier account

Featured listings

Advertising

Technician commission

Workshop subscription

Equipment rental commission

B2B procurement fees

Fleet management subscription

Maintenance management subscription

Logistics commission

Enterprise services

The final model should be based on market evidence rather than assumptions.

37. CUSTOMER TRUST RESEARCH

Research should identify what makes customers trust or distrust online parts suppliers.

Important questions include

How do customers determine whether a supplier is genuine?

How important is supplier location?

How important is physical shop verification?

How important are reviews?

How important is OEM information?

How important is product packaging?

How important is warranty?

How important is return policy?

How important is delivery time?

How important is price?

These findings will directly influence the UI/UX.

38. SUPPLIER TRUST RESEARCH

We must also understand supplier concerns.

Potential concerns include

Fraudulent customers

Payment delays

Fake orders

Returns

Platform commission

Inventory management

Customer disputes

Competitor visibility

Price competition

Data privacy

Delivery problems

Supplier onboarding complexity

These concerns must influence the business model and platform design.

39. TECHNICIAN BUSINESS RESEARCH

Research should establish

How mechanics currently find customers

How customers find mechanics

How technicians charge

How emergency services work

How technicians prove expertise

How workshops manage technicians

How customers assess technician quality

What information technicians need before accepting a job

This will determine the future technician marketplace.

40. FLEET OPERATOR RESEARCH

Fleet operators should be researched separately.

Important areas include

Number of vehicles

Maintenance processes

Parts procurement

Workshop relationships

Service schedules

Vehicle downtime

Maintenance costs

Driver management

Parts inventory

Approval processes

Procurement

Reporting

The fleet management product should be based on these real workflows.

41. PROCUREMENT RESEARCH

B2B procurement research should investigate

Who makes purchasing decisions

How suppliers are selected

How quotations are requested

How quotations are compared

Purchase orders

Approvals

Payment processes

Delivery

Invoices

Documentation

Supplier qualification

Large quantity purchases

Recurring purchases

This will determine the eventual procurement experience.

42. REGULATORY RESEARCH

The project requires a dedicated regulatory review covering

Business registration

Consumer protection

Data protection

Electronic commerce

Payments

Tax

Product standards

Product certification

Importation

Automotive parts

Equipment

Industrial products

Logistics

Equipment rental

Technician services

Marketplace liability

The regulatory review must be updated before commercial launch because requirements can change.

43. BUSINESS VERIFICATION REQUIREMENTS

MECHSOURCE should establish different verification levels.

Possible levels

Basic account

Business submitted

Identity verified

Business verified

Supplier verified

Premium or trusted supplier

Technician verified

Workshop verified

The exact naming and requirements will be determined after research.

44. PRODUCT VERIFICATION MODEL

Products may have different verification states.

Possible states

Unverified

Supplier supplied

Document verified

Manufacturer verified

Technician verified

Platform verified

AI suggested

The platform should never imply that AI identification is equivalent to manufacturer verification.

45. CUSTOMER RESEARCH

Before final UI/UX design, research should include actual potential users.

Potential research participants

Vehicle owners

Truck owners

Mechanics

Parts dealers

Workshop owners

Fleet managers

Construction businesses

Industrial businesses

Procurement officers

Equipment operators

Questions should focus on actual behaviour rather than asking users whether they "like" the idea.

46. USER INTERVIEW OBJECTIVES

Research should establish

How users currently find parts

How long it takes

Where they search

Who they ask

What information they provide

What information they do not know

How often they buy parts

What causes wrong purchases

How they compare prices

How they select suppliers

How they arrange delivery

How they handle returns

How they find mechanics

What causes delays

What information they would trust from MECHSOURCE

47. PILOT MARKET RESEARCH

Before attempting a large launch, MECHSOURCE should consider collecting real supplier and customer information in the initial markets.

Initial locations

Lagos

Ogun

Abuja

The objective is to establish

Supplier density

Demand

Part availability

Price differences

Delivery times

Common searches

Common vehicle types

Common part categories

Customer problems

Supplier behaviour

This information can influence the final product design.

48. INITIAL DATA COLLECTION STRATEGY

Data collection should be divided into

Core reference data

Supplier data
Inventory data
Customer generated data
Technician data
Workshop data
Transaction data
AI data
Maintenance data
Analytics data

Each type should have its own owner and update process.

49. DATA OWNERSHIP

Every major dataset must have a defined ownership model.

For example

MECHSOURCE owned data
Supplier owned data
Customer supplied data
Licensed third party data
Manufacturer supplied data
Public sector data
AI generated data
Technician verified data

This distinction is necessary for legal, technical and operational reasons.

50. DATA VALIDATION

Before important information becomes trusted platform information, MECHSOURCE should determine

Who validates it

What evidence is required

How often it is reviewed

What happens when information conflicts

How corrections are handled

How outdated information is marked

How incorrect supplier information is handled

51. BUSINESS OPERATIONS

MECHSOURCE will eventually require operational teams or processes for

Supplier onboarding

Supplier verification

Product moderation

Data management

Compatibility verification

Customer support

Dispute resolution

Fraud monitoring

Technician verification

Workshop verification

AI review

Content moderation

Payment reconciliation

Logistics coordination

Quality control

52. SUPPORT OPERATIONS

Customer support must eventually be able to handle

Wrong part

Missing part

Damaged part

Incorrect compatibility

Supplier dispute

Payment issue

Delivery delay

Failed delivery

Refund

Return

Technician dispute

AI identification error

Account problem

The support workflow should be designed before launch.

53. MARKETPLACE DISPUTES

The platform should establish rules for

Customer versus supplier

Customer versus technician

Customer versus workshop

Supplier versus customer

Supplier versus logistics provider

The system should record evidence such as

Order

Payment

Messages

Delivery information

Photographs

Documents

Reviews

This will help with dispute resolution.

54. DATA SECURITY REQUIREMENTS

Research must determine how sensitive information should be handled.

Potential sensitive information includes

Customer identity

Contact information

Location

Business information

Payment information

Vehicle information

Images

Messages

Documents

Supplier information

Technician information

The final technical architecture must incorporate appropriate security controls.

55. EXTERNAL INTEGRATIONS TO RESEARCH

Potential integrations include

Payment providers

Maps

Geolocation

Logistics providers

SMS

Email

Push notifications

AI services

Vehicle data providers

Parts databases

Business verification

Analytics

Cloud storage

Customer support

Accounting

Inventory systems

Enterprise procurement systems

The objective is to determine which integrations are actually necessary and which can be added later.

56. DATASET PRIORITY

Not every dataset needs to exist before MECHSOURCE begins.

The datasets should be classified as

Critical before design

Critical before launch

Required for expansion

Optional

Future research

Examples of potentially critical datasets include

Vehicle reference data

Part categories

Core part information

Compatibility data

Supplier data

Location data

Product pricing

Inventory information

These priorities will be confirmed after research.

57. DATASET DEVELOPMENT STRATEGY

MECHSOURCE should not attempt to collect every vehicle, part and machine in the world before launching.

Instead, the platform can establish a controlled initial coverage strategy.

For example

Priority vehicle brands

Priority vehicle models

Priority engine types

Priority part categories

Priority equipment

Priority locations

Coverage can then expand according to actual demand.

This allows the database to grow systematically without compromising the long term architecture.

58. AI DATA STRATEGY

AI should initially rely on a combination of

Structured MECHSOURCE data

Licensed data

Verified supplier information

Manufacturer information where available

User supplied information

Human verification

Appropriate AI models

The objective is to avoid building an AI system that produces plausible sounding but unreliable part information.

59. BUSINESS PARTNERSHIP RESEARCH

Potential partnerships may include

Parts manufacturers

Distributors

Vehicle dealers

Equipment dealers

Workshops

Technicians

Fleet companies

Logistics companies

Payment providers

Insurance related businesses where appropriate

Data providers

Industry associations

Training organisations

The partnership strategy should be investigated after the core market research.

60. RESEARCH OUTPUT

At the end of Document 02 research, MECHSOURCE should have a clear evidence based understanding of

1. Target market
2. Customers
3. Suppliers
4. Technicians
5. Workshops
6. Vehicles
7. Equipment
8. Parts
9. Compatibility
10. Data sources
11. Data licensing
12. Pricing
13. Inventory
14. Logistics
15. Payments
16. Regulations
17. Consumer protection
18. Data protection
19. Competitors
20. Business model
21. AI requirements
22. Verification requirements
23. Operational requirements
24. External integrations
25. Expansion requirements

61. RESEARCH DELIVERABLES

The research stage should ultimately produce

Market Research Report

Competitor Analysis

Customer Research

Supplier Research

Technician and Workshop Research

Vehicle Data Source Assessment

Equipment Data Source Assessment

Parts Data Source Assessment

Compatibility Data Strategy

OEM and Cross Reference Data Strategy

AI Dataset Strategy

Data Licensing Assessment

Supplier Verification Model

Product Verification Model

Logistics Research

Payment Research

Regulatory Research

Business Model Research

Initial Market Coverage Strategy

Data Governance Strategy

62. IMPORTANT DECISION

MECHSOURCE must not build its core product around an assumption that there is one perfect publicly available dataset containing every vehicle, engine, OEM number and compatible part.

There may be multiple data sources, different licensing arrangements, incomplete coverage and conflicting information.

Therefore, MECHSOURCE needs its own structured data strategy.

The platform should be capable of combining

Licensed reference data

Manufacturer information

Verified supplier information

Technician knowledge

Customer contributions

Platform generated data

AI assistance

Human verification

63. RELATIONSHIP BETWEEN DOCUMENT 01 AND DOCUMENT 02

Document 01 defines

WHAT MECHSOURCE IS.

Document 02 defines

WHAT MECHSOURCE NEEDS FROM THE REAL WORLD TO WORK.

The two documents must remain connected.

Document 02 should not change the fundamental product vision without documenting why.

If research reveals that a feature is impractical, too expensive, legally restricted or dependent on unavailable data, the requirement should be flagged for review rather than silently removed.

64. OUTPUT REQUIRED BEFORE DOCUMENT 03

Before we move to the UX/UI Specification, we need to know

What users need

What users actually do

What data exists

What data does not exist

What data can legally be used

What data must be collected

What data must be verified

What information changes frequently

What suppliers can provide

What customers need to provide

What external services are required

What regulatory constraints affect the product

What information must appear in the interface

What information should remain behind the scenes

Which features are ready for design

Which features require further validation

65. FINAL DATA PRINCIPLE

The quality of MECHSOURCE will depend heavily on the quality of its underlying information.

A beautiful interface cannot compensate for incorrect compatibility data.

An advanced AI cannot compensate for poor reference data.

A large marketplace cannot compensate for unverified suppliers.

Therefore

DATA QUALITY + VERIFICATION + USER EXPERIENCE + TRUST + TECHNOLOGY

must work together.

This principle should guide all subsequent MECHSOURCE research, UI/UX decisions and technical architecture.